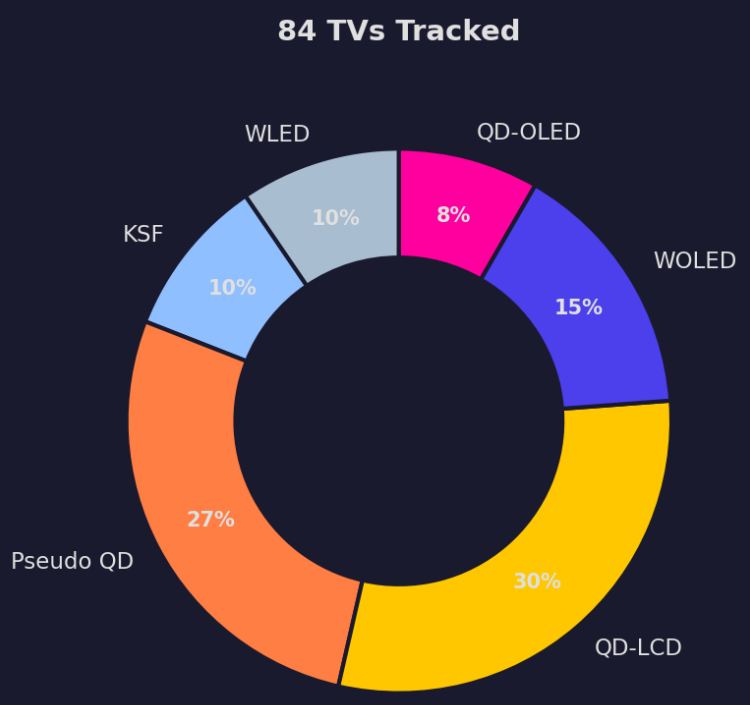
Display Technology Intelligence Report

April 2026

84 4K TVs tracked | 71 with pricing | 6 technologies | 3 8K excluded



1. Display Technology Overview

Section 1: Display Technology Overview -- April 2026

1.1 Catalog Stability

The April 2026 tracked universe holds steady at 84 televisions across six distinct display technologies, with zero additions and zero removals this month. That kind of stasis is relatively rare in a category that typically sees mid-cycle refreshes and inventory churn through Q1, and it reflects a market that has largely finished digesting the CES 2025 launch wave. Three 8K models remain excluded from scoring aggregates to prevent panel resolution from confounding technology-level comparisons -- a methodological carve-out that keeps the performance benchmarks clean and directly comparable across the mainstream 4K segment.

The technology distribution skews heavily toward LCD-based solutions: QD-LCD leads the count at 25 units, followed by Pseudo QD at 23 and WLED at 8, meaning LCD variants account for 56 of the 84 tracked sets. Emissive technologies -- WOLED at 13 units and QD-OLED at 7 -- together represent just 24% of the catalog despite commanding the top performance tiers. KSF rounds out the field at 8 units, occupying the mid-market LCD segment alongside Pseudo QD.

The stable catalog is a useful baseline for what follows: with no new entrants diluting or inflating technology-level averages, the scores this month represent the most settled, apples-to-apples snapshot the dataset has produced in several reporting cycles.

1.2 Performance Snapshot

QD-OLED sits at the apex of every scored use case. Its mixed-usage mean of 8.6, home theater mean of 8.9, and gaming mean of 8.9 are the highest of any technology in the dataset -- and the floor of its range tells the more impressive story: the lowest-scoring QD-OLED in home theater still posts an 8.6, meaning the technology has effectively eliminated weak performers from its tracked cohort. WOLED trails by a narrow margin, averaging 8.5 in home theater and 8.6 in gaming, but it spans a wider performance range -- its mixed-usage floor drops to 7.2, compared to QD-OLED's 8.2 minimum.

The gap widens sharply below the OLED tier. QD-LCD averages 7.9 in mixed usage and 8.0 in home theater, making it the strongest LCD contender and a credible alternative for buyers who cannot justify the OLED premium. But the distance between QD-LCD's ceiling (8.7 in home theater) and QD-OLED's floor (8.6) illustrates how compressed the top of the market has become. Below QD-LCD, the drop is abrupt: KSF and Pseudo QD both average 6.5 in mixed usage, and WLED trails the field at a mean of just 5.2 -- a full 3.4 points behind QD-OLED on the same metric.

The practical implication is a market stratified into three tiers with little overlap: a premium OLED tier averaging above 8.5, a capable QD-LCD mid-tier in the high 7s-to-low 8s, and a crowded entry tier where KSF, Pseudo QD, and WLED compete largely on price rather than performance differentiation.

1.3 FWHM Spectral Analysis

Full Width at Half Maximum (FWHM) measures how narrow -- and therefore how pure -- a display's color emission peaks are. Narrower peaks mean less spectral overlap between primaries, which translates directly to wider color gamut coverage and more accurate saturation. QD-OLED produces the tightest green emission in the dataset at a mean of 27.8 nm, edging out QD-LCD's 29.6 nm green FWHM. Both are dramatically narrower than WOLED's 87.8 nm green peak -- a figure that reveals why WOLED, despite its outstanding contrast performance, requires heavier color processing to reach competitive gamut coverage.

On the red channel, KSF posts the narrowest peak at just 9.1 nm, a function of the phosphor's quantum-dot-like emission properties. This makes KSF spectrally efficient on red despite its relatively modest overall scores -- a reminder that narrow FWHM is necessary but not sufficient for top performance. QD-LCD's red FWHM of 29.8 nm and QD-OLED's 37.1 nm are both competitive, while WLED's red peak at 78.7 nm explains much of that technology's gamut limitations alongside its broad 70.9 nm green.

The spectral data reinforces the performance hierarchy: the two technologies with the narrowest combined green and red peaks -- QD-OLED and QD-LCD -- occupy the top two performance tiers by score, while WOLED compensates for its wide emission profile through perfect per-pixel contrast. WLED's broad peaks on both channels leave it structurally disadvantaged on color quality regardless of backlight intensity improvements.

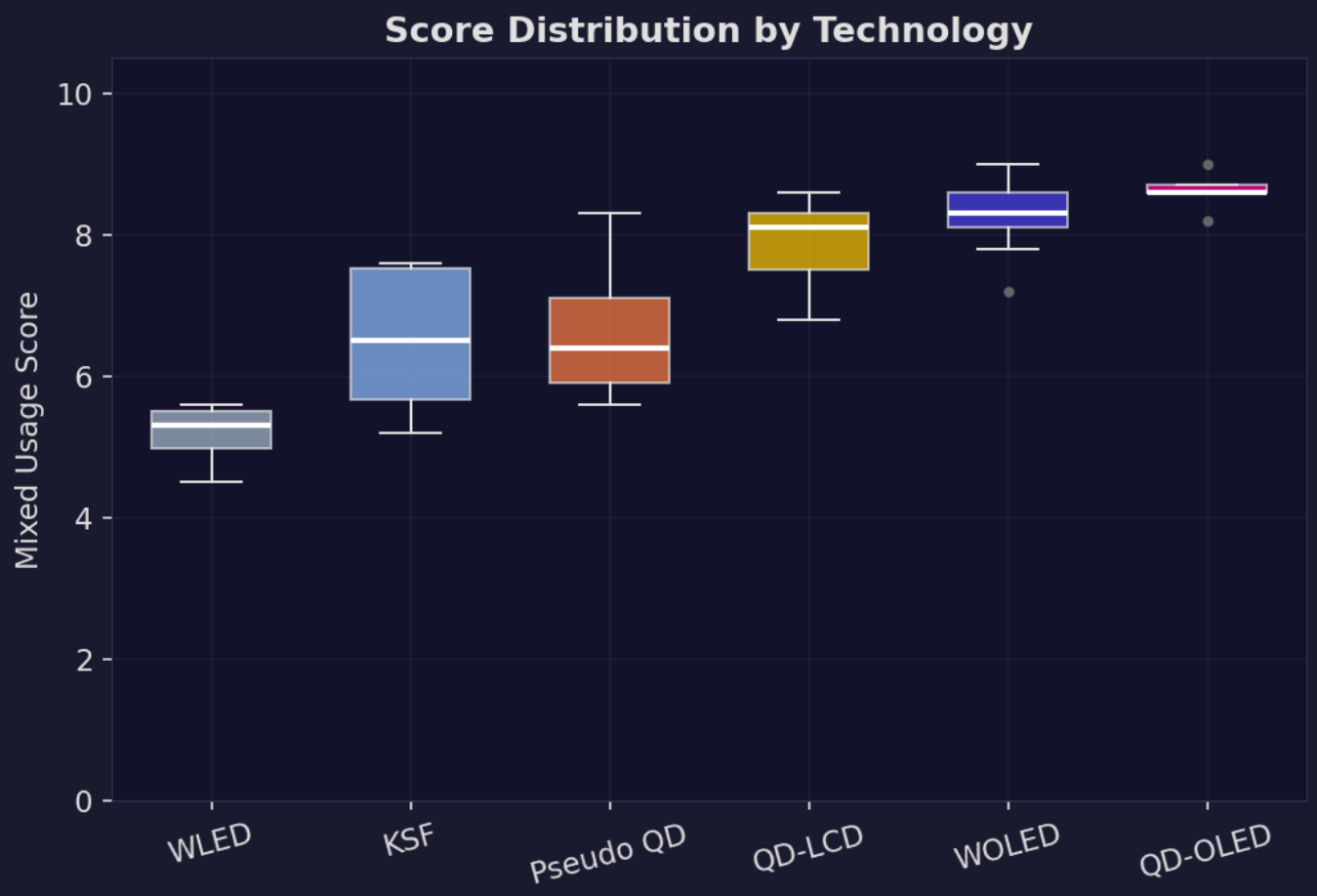
1.4 Model Year Composition

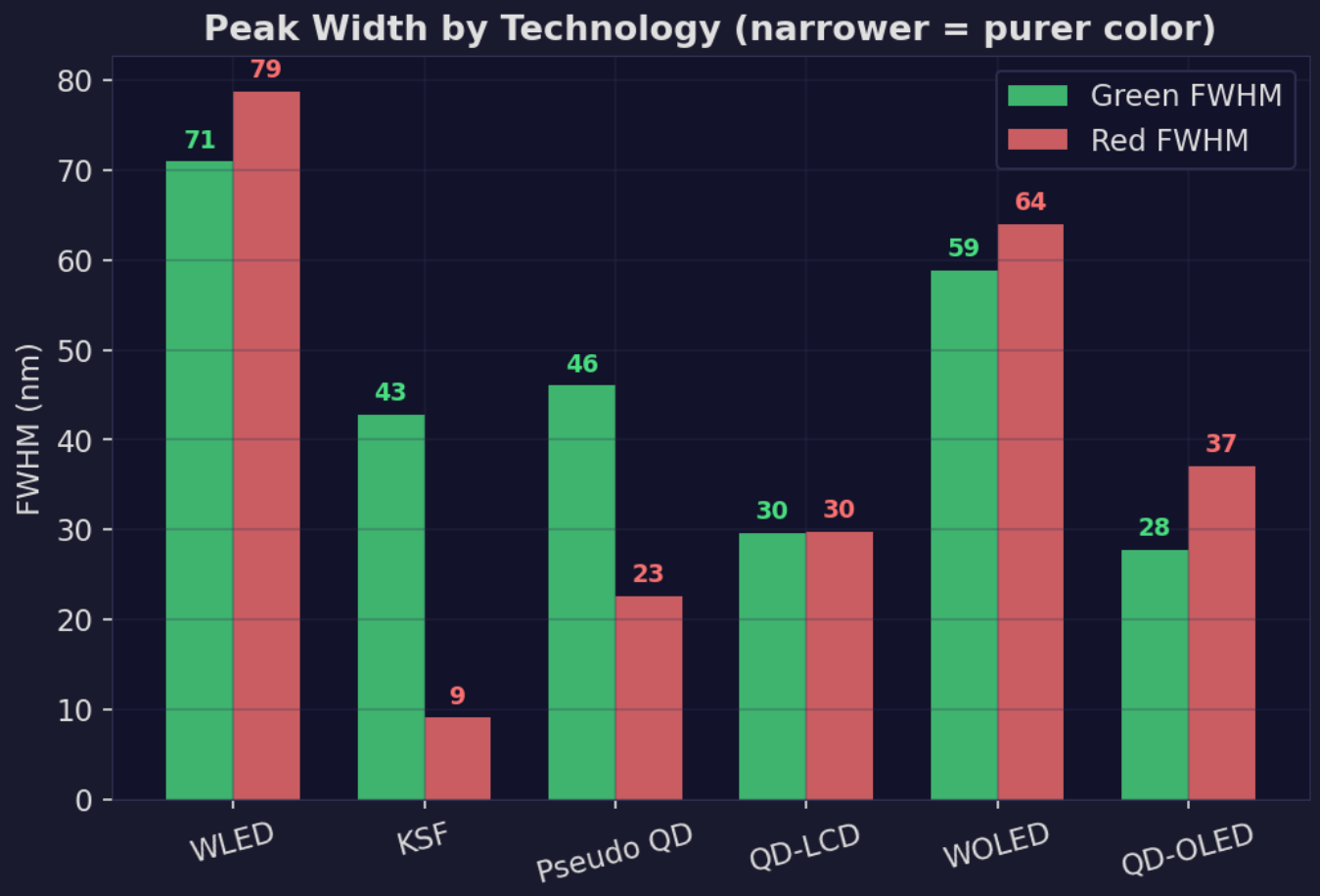
The dataset does not yet contain confirmed 2026-model-year units in sufficient quantity to produce a statistically meaningful year-over-year comparison -- a gap that will close as spring retail launches are logged in coming months. What the current data does establish is the 2025-model baseline: 41 tracked sets carry a 2025 model designation, averaging 7.3 in both mixed usage and gaming -- scores that align closely with the QD-LCD tier and confirm that the 2025 wave skewed toward the upper-mid market rather than either the entry or premium extremes.

The 43 non-2025 units in the dataset are primarily older model-year sets that have retained retail presence through carry-over pricing and inventory dynamics. Their continued inclusion keeps the catalog representative of what consumers actually encounter on shelves, where a 2023-vintage QD-LCD frequently competes alongside a 2025 WOLED at overlapping price points.

As 2026 models enter the tracking universe -- expected to accelerate through May and June -- the

year-over-year comparison will become the report's most actionable data point; watch for whether new entrants shift QD-OLED's floor score upward or whether WOLED's gaming figures close




Peak Width by Technology (narrower = purer color)
FWHM (nm)
0
10
20
30
40
50
60
70
80
Green FWHM
Red FWHM
71
79
43
9
46
23
30
30
59
64
28
37
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED

2. Pricing Intelligence

Section 2: Pricing Intelligence

2.1 Current Pricing Landscape

Across 71 priced models in this month's dataset, retail prices span an enormous range -- from entry WLED sets under $300 to flagship QD-OLED panels pushing well past $3,000. The median price by technology tells that story cleanly: WLED sits at $436, while QD-OLED commands a $2,338 median, a gap of over $1,900 between the floor and ceiling of the market. The middle of the stack clusters around Pseudo QD at $725, WOLED at $1,600, and QD-LCD at $1,909, with KSF at $1,280 occupying the transition zone between mass-market and premium LCD.

The per-square-meter lens sharpens the competitive picture considerably. WLED lands at $565/m2, establishing the commodity baseline. From there, each technology tier commands a meaningful step-up: Pseudo QD runs $886/m2 (+57% vs. WLED), KSF reaches $1,144/m2 (+102%), QD-LCD climbs to $2,208/m2 (+291%), WOLED to $2,041/m2 (+261%), and QD-OLED tops the stack at $2,568/m2 -- a 355% premium over standard WLED on a screen-area-adjusted basis. That 355% figure is the most important single number in the pricing landscape: it quantifies exactly how much consumers are paying for the Samsung and Sony QD-OLED experience relative to the mass-market alternative.

Retailers appear to have drawn firm technology boundaries that align with the manufacturing cost structure -- there is almost no price overlap between the LCD emissive tiers (WLED, Pseudo QD, KSF) and the premium display tiers (QD-LCD, WOLED, QD-OLED), which suggests brand pricing discipline rather than a market in active disruption. Whether that discipline holds through the summer promotional cycle will determine whether OLED volumes move meaningfully.

2.2 Price Trends

Eight weeks of pricing data provide a credible short-run trend baseline, though single-season reads should be treated as directional rather than definitive. The most significant move belongs to WLED, which rose 8.2% on a per-m2 basis -- from $522 to $565/m2 -- the largest percentage shift of any technology this period. That upward drift in the commodity tier is counterintuitive heading into spring and warrants close monitoring; if it reflects panel cost pass-through rather than promotional timing, margin compression at budget brands becomes a real risk.

The premium tiers held almost completely flat. WOLED did not move at all ($2,041/m2 in both periods), and QD-OLED slipped just 0.7%, from $2,586 to $2,568/m2 -- essentially noise. QD-LCD edged up 0.7% and

KSF rose 3.8%, while Pseudo QD was the lone decliner in the LCD stack, dropping a negligible 0.3%. The broad takeaway: premium display pricing is sticky, and manufacturers are not discounting to drive volume in early Q2.

The combination of rising WLED prices and flat OLED prices is quietly compressing the premium gap on a percentage basis, even if absolute dollar distances remain wide. A consumer comparing a $436 WLED to a $1,600 WOLED faces a smaller relative hurdle than they did eight weeks ago -- and that dynamic, if it continues, could shift the value calculus for mid-market shoppers by mid-year.

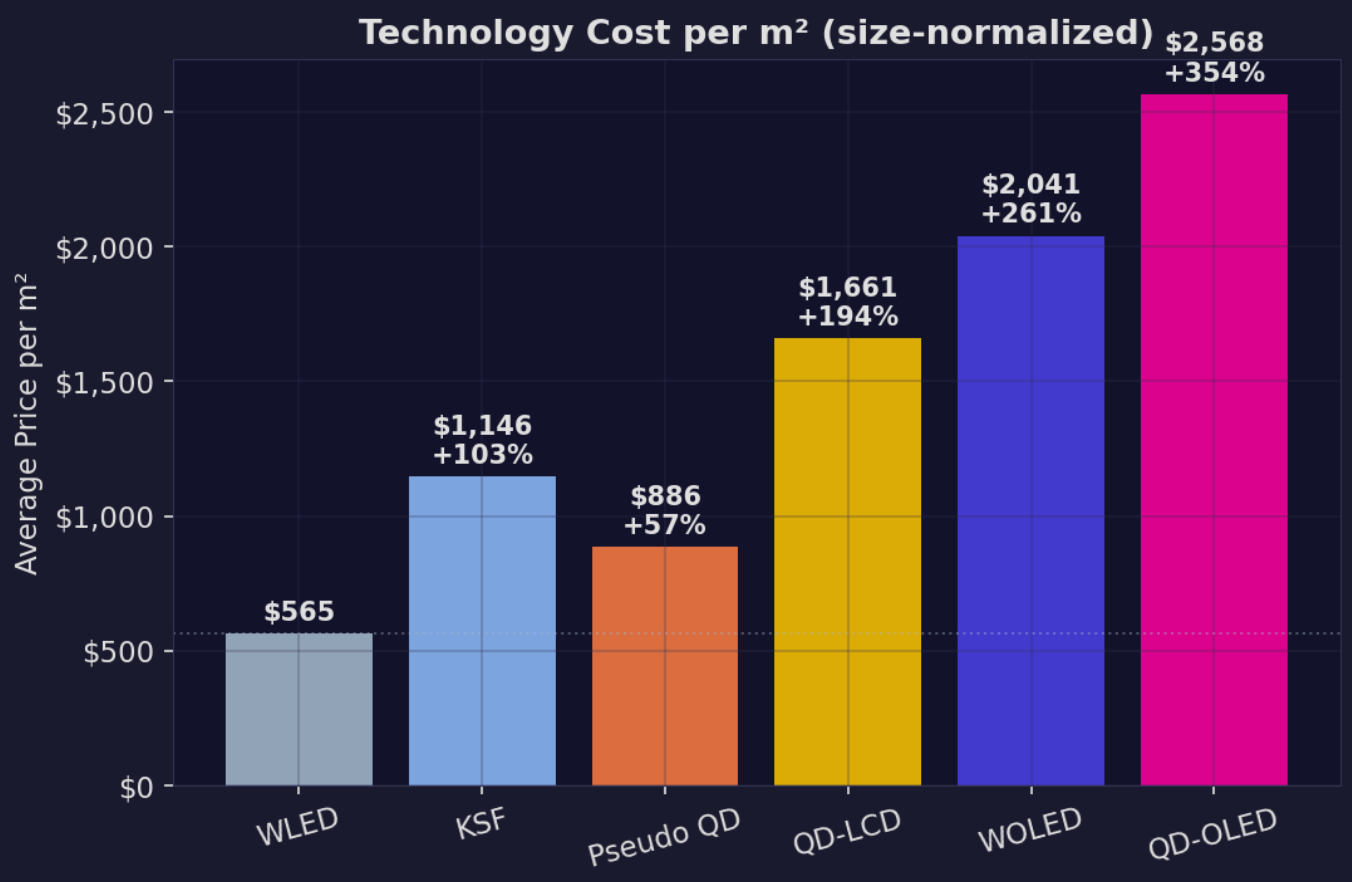
2.3 Value Analysis

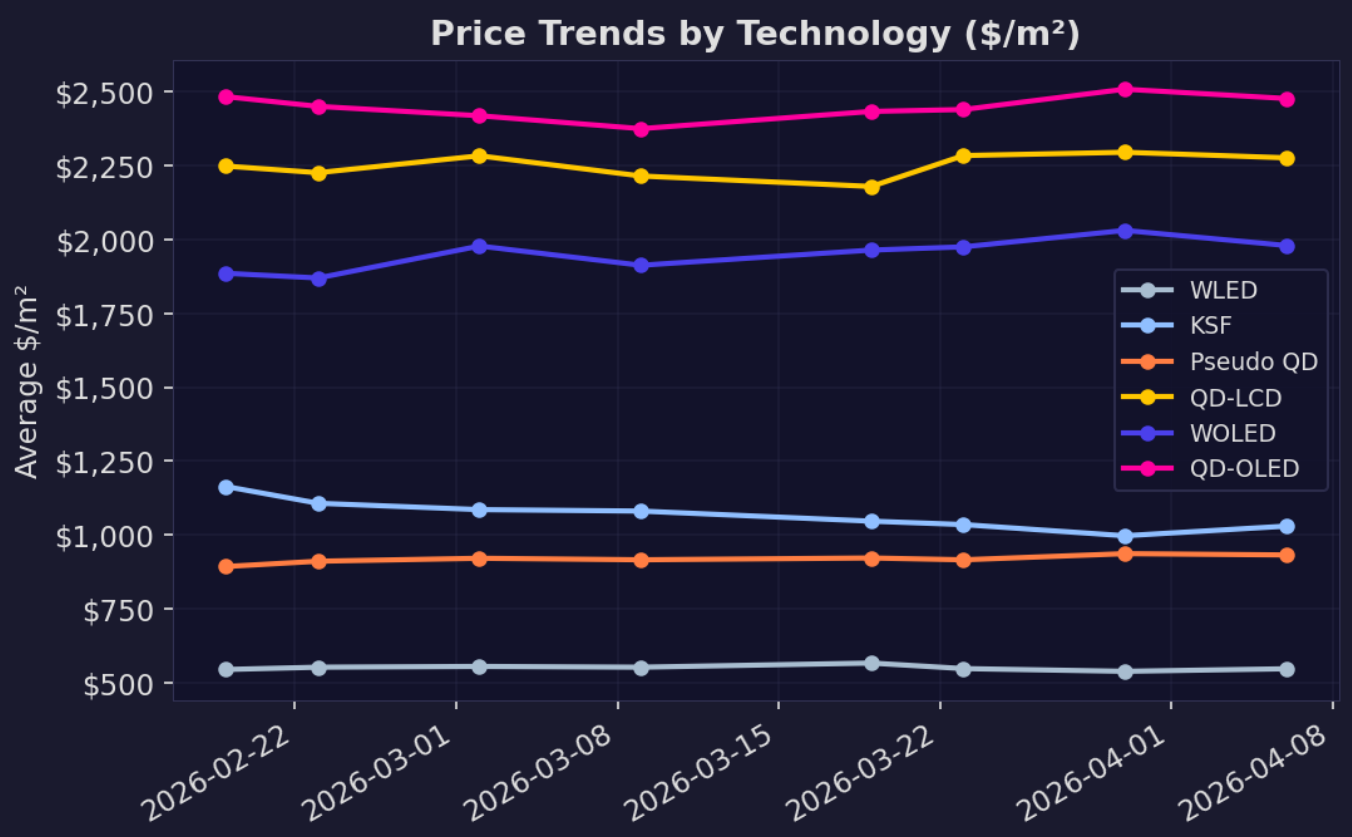
The dollar-per-performance-point metric cuts through sticker price and identifies where the market genuinely rewards buyers. The top value positions in this month's data are tightly clustered: the Hisense QD6QF (Pseudo QD, $370, mixed-usage score 5.9) and the TCL S551G (KSF, $370, score 5.9) both land at exactly $63 per performance point, tying for the top value spot. The Samsung U7900F (WLED, $330, score 5.2) also hits $63/point, confirming that the entry tier can compete on efficiency even with lower absolute scores. Just behind them, the Hisense QD7QF (Pseudo QD, $438, score 6.7) comes in at $65/point, and the Hisense U6N (Pseudo QD, $451, score 6.7) rounds out the top five at $67/point.

The pattern across those top five is striking: three of the five are Hisense models, all in the Pseudo QD tier, and none exceed $451. Hisense has engineered a value density that neither Samsung's WLED nor TCL's KSF can consistently replicate across the lineup -- the brand is systematically occupying the sweet spot where performance-per-dollar peaks. The TCL S551G is the notable exception, proving KSF can compete on value terms when priced aggressively, but TCL's broader KSF lineup does not sustain that efficiency at scale given a median KSF price of $1,280.

No OLED model appears in the top value tier. At $2,568/m2 for QD-OLED and $2,041/m2 for WOLED, the premium display segment is pricing for experience and brand cachet, not performance-per-dollar efficiency. That is not a flaw -- it reflects a different buyer -- but it does mean the value narrative for 2026 belongs entirely to the mid-tier LCD technologies, specifically Pseudo QD and selectively priced KSF.

The brands that can hold Pseudo QD pricing near $400 - $450 while sustaining scores above 6.5 will own the largest addressable value segment heading into Q3 promotional season.


Technology Cost per m² (size-normalized)
Average Price per m²
$2,500
$2,000
$1,500
$1,000
$500
$0
$565
$1,146
+103%
$886
+57%
$1,661
+194%
$2,041
+261%
$2,568
+354%
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED


Price Trends by Technology ($/m²)
Average $/m²
$2,500
$2,250
$2,000
$1,750
$1,500
$1,250
$1,000
$750
$500
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED
2026-02-22
2026-03-01
2026-03-08
2026-03-15
2026-03-22
2026-04-01
2026-04-08

Price vs Performance
Mixed Usage Score
Price ($)
0
2
4
6
8
10
$0
$2,000
$4,000
$6,000
$8,000
$10,000
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED
Value Frontier

3. Macro & Industry Context

Section 3: Macro & Industry Context

The Display Technology Stack, April 2026

The TV market our database tracks spans 84 models across six distinct display technologies, and that taxonomy alone tells a story. QD-LCD dominates coverage at 25 models, followed closely by Pseudo QD at 23 -- together accounting for nearly 57% of tracked units. That concentration reflects the LCD ecosystem's continued stranglehold on mid-market shelf space, where manufacturers can hit aggressive price points while deploying just enough marketing language to obscure meaningful performance differences. WOLED and QD-OLED together total just 20 models, but they punch far above their numeric weight in both revenue per unit and critical attention. The six-tier technology stack we track -- from basic WLED phosphor through KSF narrow-red to true quantum dot and self-emissive panels -- maps almost perfectly onto the industry's pricing ladder, with each rung representing genuine physics, not just branding.

Samsung's 24-model footprint is the largest in the database, a direct consequence of the company's vertical integration across both QD-LCD and QD-OLED supply chains. Hisense at 15 models and TCL at 13 represent the aggressive Chinese OEM push into mid-tier QD-LCD and the murkier Pseudo QD category, where KSF phosphor technology gets rebranded with quantum-adjacent terminology to capture margin without the cost of true quantum dot film. LG's 11 models concentrate almost entirely in WOLED and select QD-LCD lines, reflecting the company's identity as the primary OLED panel supplier to the industry -- including to Sony, whose 10-model presence leans heavily on WOLED glass sourced from LG Display.

Technology Differentiation: What the Physics Actually Deliver

The gap between QD-OLED and everything else in our database is not incremental -- it is structural. QD-OLED's self-emissive architecture eliminates the backlight entirely, enabling per-pixel extinction ratios that LCD-based panels cannot approach regardless of local dimming zone count. WOLED, also self-emissive, shares that fundamental advantage but routes light through a white subpixel plus color filter stack, which costs roughly 20-30% of peak color volume compared to QD-OLED's direct quantum dot color conversion. In practice, WOLED panels -- primarily LG's B5, C5, and G5 series in our tracked universe -- deliver superior black levels and motion handling versus any LCD, while QD-OLED panels from Samsung and Sony push further into wide color gamut coverage and peak specular highlight brightness.

The Pseudo QD category, tracked at 23 models, deserves particular scrutiny. These are functionally KSF narrow-red phosphor panels sold under brand names that imply quantum dot performance -- a gap that only rigorous third-party scoring from sources like RTINGS.com exposes. True QD-LCD, by contrast, uses cadmium-free quantum dot enhancement film to achieve BT.2020 color gamut coverage in the 70-80% range, a meaningful separation from KSF's typical 60-65% ceiling. Budget WLED panels at the base of the stack are essentially commodity components, tracked in our database primarily to establish performance

baselines and model the price-to-performance curve across the full market.

Where the Industry Is Heading

Three technologies are actively pressuring the current six-tier stack. Tandem OLED -- stacking two emissive layers to double photon output without sacrificing lifespan -- entered the laptop display market in 2024 and is transitioning toward large-format TV panels. LG Display has publicly committed to tandem WOLED production for premium TV sizes, which would directly address the brightness deficit that keeps WOLED below QD-OLED on peak HDR benchmarks. MicroLED, meanwhile, remains commercially confined to 100"+ installations and Samsung's The Wall product line; at current yields and price points, it does not threaten the sub-85" mass market within a 24-month horizon. Perovskite quantum dots represent the most disruptive long-term vector -- offering theoretically narrower emission spectra than cadmium-free QD materials at potentially lower manufacturing cost -- but commercialization challenges around stability and toxicity keep perovskite outside our near-term tracking scope.

The editorial takeaway for this team is direct: the Pseudo QD category's 23-model presence in our database represents the single largest misalignment between consumer perception and actual panel performance in the current market, and the brands exploiting that gap most aggressively -- primarily Chinese OEMs competing on price -- will face the sharpest reckoning as tandem OLED drives true self-emissive pricing into the $800-$1,200 range and eliminates the LCD mid-tier's primary value proposition.

4. Year-over-Year Technology Trends

Section 4: Year-over-Year Technology Trends

4.1 Score Trajectories: Who Earned Their Stripes

The standout performer in mixed-usage score improvement between the 2024 and 2025 model years is KSF, which posted a +1.3-point gain, jumping from 6.0 to 7.3. That magnitude of single-year improvement is rare across any display category and signals meaningful advances in KSF panel tuning and processing -- not just incremental firmware polish. The caveat is sample size: the KSF cohort shrank from 5 models in 2024 to just 2 in 2025, so the score lift reflects a flight to quality within a thinning lineup rather than broad-based execution across the category.

WOLED delivered the most consistent upward movement among premium tiers, nudging from 8.2 to 8.4 -- a +0.2-point gain that, at this altitude of the scoring range, represents genuine refinement. WLED and Pseudo QD each also gained +0.2 points, though both remain in mid-tier score territory (5.2 and 6.6, respectively), where the competitive stakes are lower. The lone decliner is QD-LCD, which slipped -0.3 points to 7.7 despite holding the largest model count at 12 sets. A broader, more democratized lineup introduced more variance -- and pulled the average down.

QD-OLED held perfectly flat at 8.6, defending its position at the top of the scoring hierarchy for a second consecutive year. Flat scores at the summit can be read two ways: either the technology has hit a refinement plateau, or the 2025 cohort was curated conservatively. Either way, no rival closed the gap.

4.2 Pricing Shifts: Compression Below, Expansion Above

The most dramatic pricing story of the 2024-to-2025 transition belongs to KSF, where cost per square meter more than doubled -- rising +102% from $906/m2 to $1,831/m2. Combined with the shrinking model count, this repositioning is deliberate: manufacturers are treating KSF as a differentiated premium offering rather than a volume play. WOLED also moved upmarket, with prices climbing +29.8% to $2,367/m2, reinforcing LG Display's leverage over the only commercial large-panel OLED supply chain not tied to Samsung's QD architecture.

Meanwhile, the mid-tier is experiencing aggressive commoditization. Pseudo QD prices collapsed -31.0% to $724/m2, and QD-LCD shed -24.7% to land at $1,463/m2. These are not soft corrections -- they represent structural price erosion driven by panel oversupply and intensifying competition from Chinese manufacturers pushing screen-size-per-dollar metrics. WLED, already the entry-level anchor of the dataset, fell another -9.1% to $549/m2, continuing its multi-year compression toward commodity floor pricing.

QD-OLED held nearly flat at $2,424/m2 -- a +1.8% move that amounts to inflation-level maintenance. At that

price point, with a score of 8.6 that no other technology matched, Samsung Display is under no pressure to cut. The premium is justified by the data, and the market appears to agree.

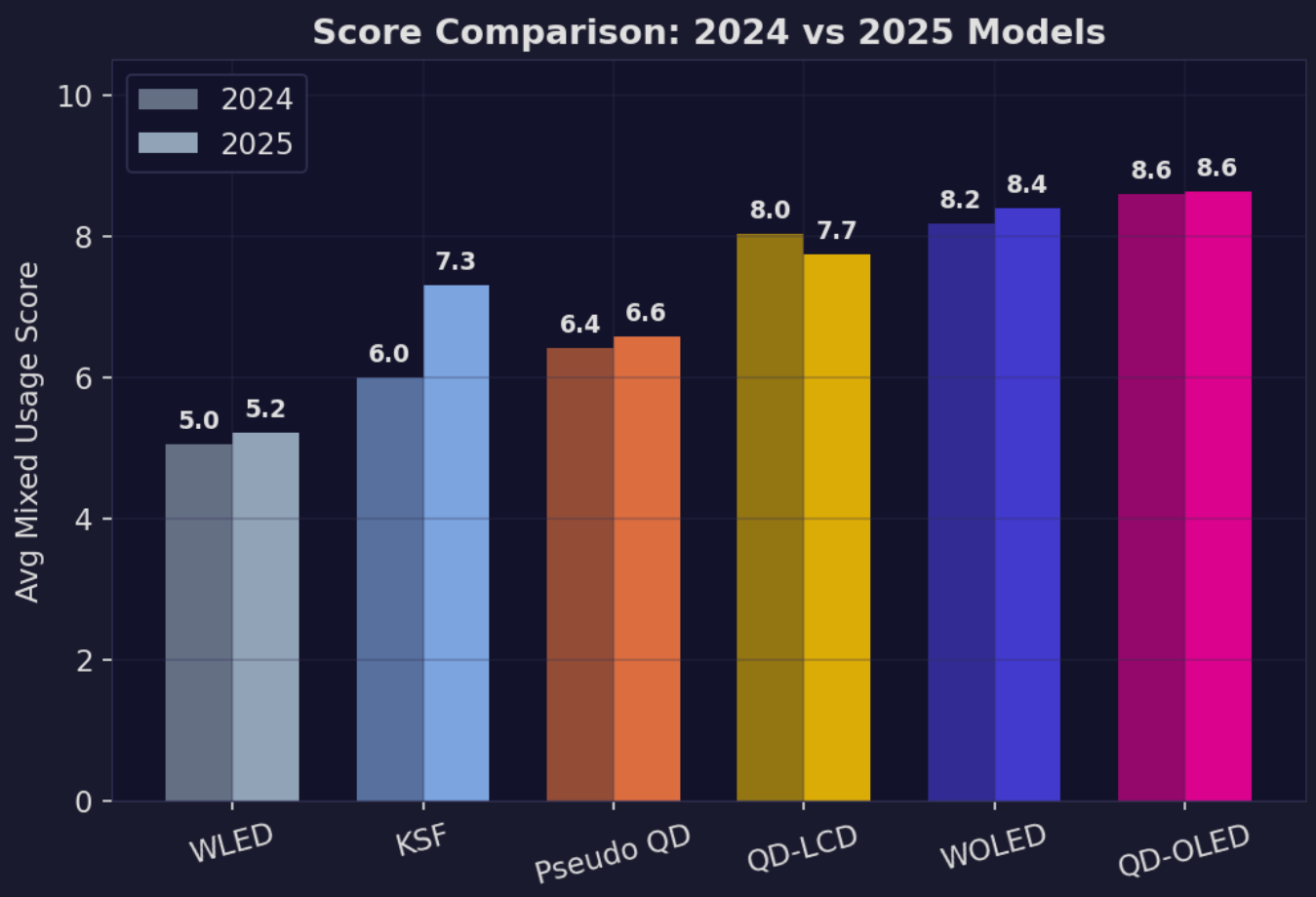
4.3 Competitive Positioning: The QD Gap Holds -- For Now

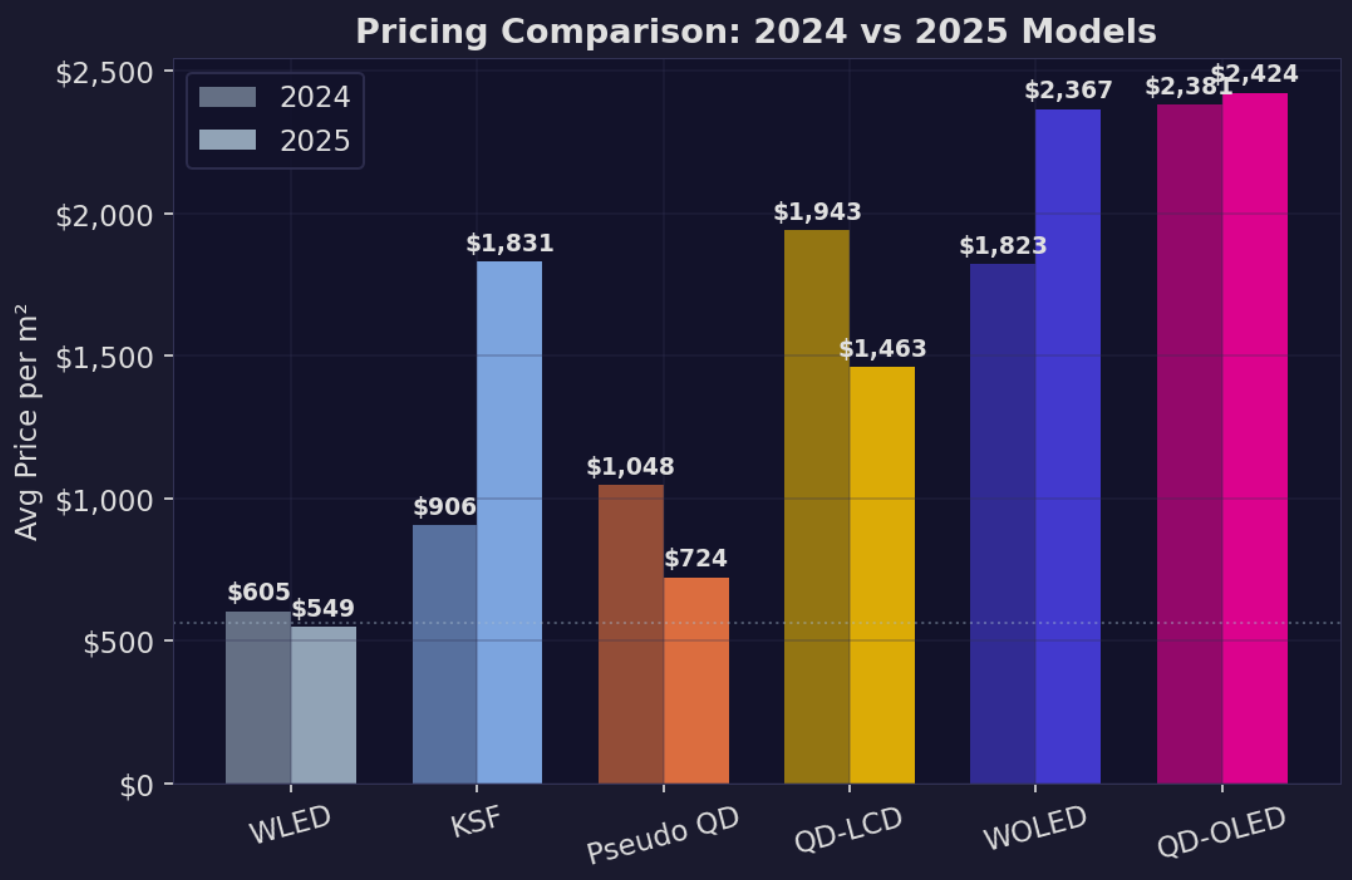
The gap between QD-class technologies and the rest of the market did not narrow in 2025 -- it calcified. QD-OLED at 8.6 and WOLED at 8.4 sit a full 0.9 points above QD-LCD at 7.7, which was itself the highest-scoring non-OLED technology. That spread was comparable in 2024, meaning a year of product releases and engineering investment produced no structural convergence. Pseudo QD at 6.6 remains anchored in a completely different competitive tier despite its -31% price cut, suggesting discounting alone cannot substitute for panel architecture advantages.

KSF's score surge to 7.3 is the one credible challenge to the LCD-to-OLED stratification narrative. If its improvement trajectory continues into 2026 model years -- and if manufacturers commit to more than two flagship models -- KSF could plausibly compete with QD-LCD on performance while undercutting it on price. That scenario remains speculative given current volume signals, but the score data makes it technically plausible for the first time.

The divergence between WOLED's price increase and QD-LCD's price decline is reshaping the value proposition calculus: a buyer choosing WOLED over QD-LCD in 2025 pays a $904/m2 premium for a +0.7-point score advantage, versus a $438/m2 premium in 2024 -- a gap that widened significantly in a single year.

So what: With QD-OLED scores frozen at the top, WOLED prices rising, and mid-tier LCD categories in free fall, the 2026 model year will force manufacturers into a binary bet -- either close the performance gap through panel innovation or race to the bottom on cost, because the middle ground is vanishing fastest.


Score Comparison: 2024 vs 2025 Models
2024
2025
Avg Mixed Usage Score
0
2
4
6
8
10
5.0
5.2
6.0
7.3
6.4
6.6
8.0
7.7
8.2
8.4
8.6
8.6
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED


Pricing Comparison: 2024 vs 2025 Models
2024
2025
Avg Price per m²
$2,500
$2,000
$1,500
$1,000
$500
$0
$605
$549
$906
$1,831
$1,048
$724
$1,943
$1,463
$1,823
$2,367
$2,381
$2,424
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED